

using two vacuum tubes as the active (amplifying) elements for each bit of information storage. Later implementations using bipolar transistors could operate at up to 20 million state transitions per second as early as 1963.”

State transitions mean what the term sounds like, and naturally, changes in state are essential to information processing and flow. Besides, that excerpt highlights that the flip-flop used the vacuum tube (which de Forest introduced), and the comparison with transistors (we’ll come to that bit in a bit) is illustrative of exponential progress.



The other William Henry!

For information on flip-flops (you’ll need to at least know about logic gates), visit <http://wearcam.org/ece385/lectureflipflops/flipflops/>.

Off-topic, we should quote this from the people’s encyclopaedia of Web 2.0 fame: “Frank Wilfred Jordan invented together with William Henry Eccles the so called “flip-flop” circuit in 1919. There is little else known about him.” Indeed, try Googling him!

1.7 IBM

1924

Several mergers had resulted in the Computing-Tabulating-Recording Company; this was renamed International Business Machines Corporation in 1924 under Thomas J Watson Sr. as president. By that time, the C-T-R Company’s business had expanded in terms of location as well as function; they already had three manufacturing facilities in Europe.

A little known fact is that apart from working at the forefront of computer technology, IBM was also involved in social reform